DATA STRUCTURE

LAB PROGRAMS

7. Write a C program to implement Array operations such as Insert, Delete and Display

```
#include <stdio.h>

int main() {
    int arr[100], n, i, pos, value;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Insert Operation
    printf("Enter position to insert: ");
    scanf("%d", &pos);

    printf("Enter value to insert: ");
    scanf("%d", &value);
```

```
for(i = n; i >= pos; i--) {
    arr[i] = arr[i - 1];
}

arr[pos - 1] = value;
n++;

printf("\nArray after insertion:\n");
for(i = 0; i < n; i++) {
    printf("%d ", arr[i]);
}

// Delete Operation
printf("\n\nEnter position to delete: ");
scanf("%d", &pos);

for(i = pos - 1; i < n - 1; i++) {
    arr[i] = arr[i + 1];
}

n--;

// Display Operation
printf("\nArray after deletion:\n");
for(i = 0; i < n; i++) {
```

```
        printf("%d ", arr[i]);
    }

    return 0;
}
```

```
Enter number of elements: 6
Enter array elements:
4 5 6 77  88 5
Enter position to insert: 3
Enter value to insert: 55

Array after insertion:
4 5 55 6 77 88 5

Enter position to delete: 2

Array after deletion:
4 55 6 77 88 5
--------------------------------
Process exited after 22.87 seconds with return value 0
Press any key to continue . . .
```